

Purification and Analysis of Organic Compounds

SHORT NOTES

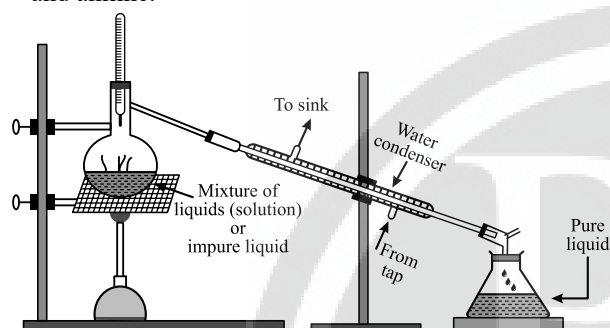
Crystallisation: It is based on the difference in the solubilities of the compound and the impurities in a suitable solvent.

Fractional Crystallization: Difference in solubilities of organic compounds and impurities are very less.

eg. urea and copper sulphate.

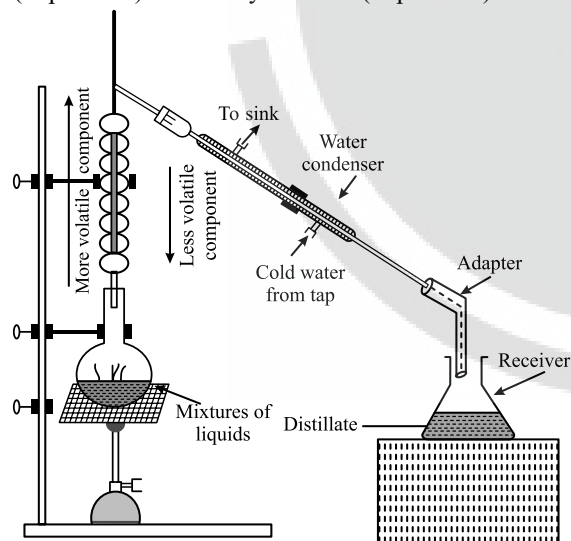
Distillation

- (i) **Simple distillation:** Difference in boiling points of compounds is more than 40°C . e.g. chloroform (b. p. 334K) and aniline.



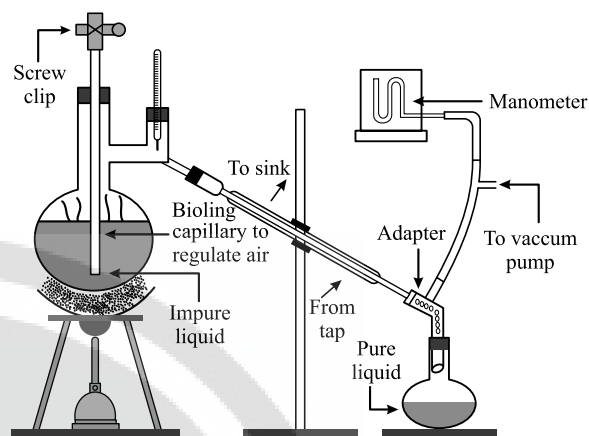
Simple distillation

- (ii) **Fractional distillation:** Difference in boiling points of compounds is very less (5°C to 10°C). e.g. - acetone (b. p. 329K) and methyl alcohol (b. p. 338K).



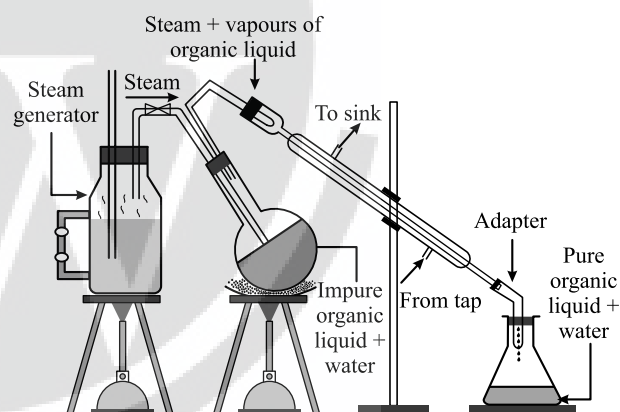
Fractional distillation

- (iii) **Vacuum distillation:** Used for organic compounds which decompose at or below their boiling points. e.g. Glycerol.



Distillation under reduced pressure

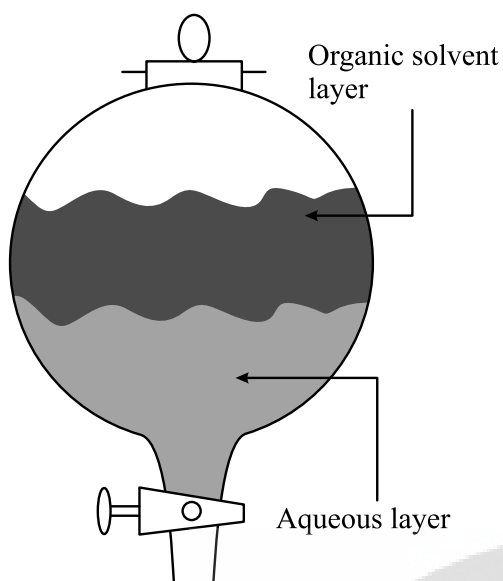
- (iv) **Steam distillation:** Used for organic compounds which are immiscible with water and are steam volatile. e.g. Aniline.



Apparatus for steam distillation

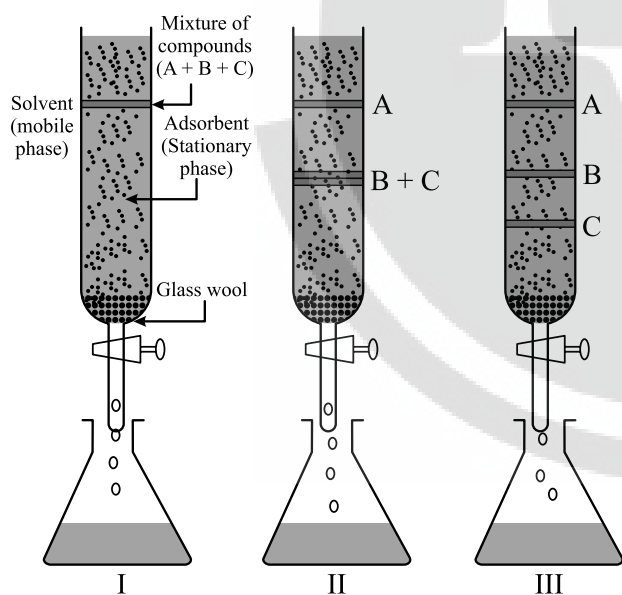
Differential extraction: Used to extract pure organic compounds from their aqueous solution by shaking with organic solvent in which they are highly soluble.

e.g. Benzoic acid from its aqueous solution using benzene.



Chromatography

- ❖ Used to purify small samples.
- ❖ Based on selective adsorption or partition between stationary and mobile phase.
- ❖ Column Chromatography based on adsorption, used for bulk quantities.
- ❖ Thin layer Chromatography based on adsorption, used for quantitative analysis.



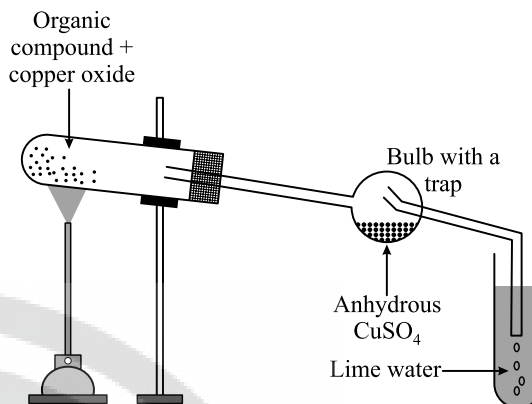
Column chromatography-stage I, II and III represent the progressive separation of the mixture into three bands

- ❖ Paper Chromatography- based on partition and used for quantitative and qualitative analysis.

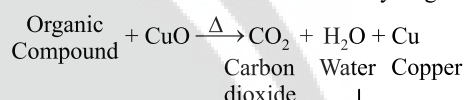
QUALITATIVE ANALYSIS

Detection of C, H, N, halogens P, S, and oxygen.

Detection of Carbon and Hydrogen



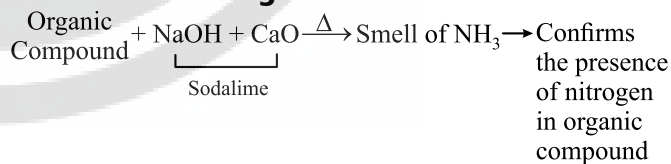
Detection of carbon and hydrogen



Passed into lime water Ca(OH)_2
 Lime water turns milky
 Carbon (C) present in organic compound

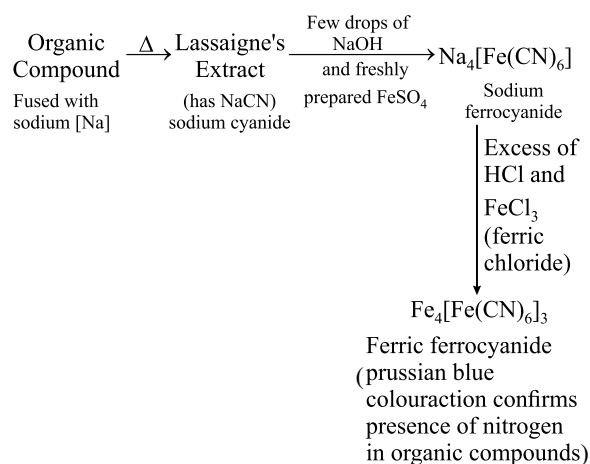
Passed into anhydrous CuSO_4
 Copper Sulphate turns blue
 Hydrogen (H) present in Organic Compound

Detection of Nitrogen



Lassaigne's Extract

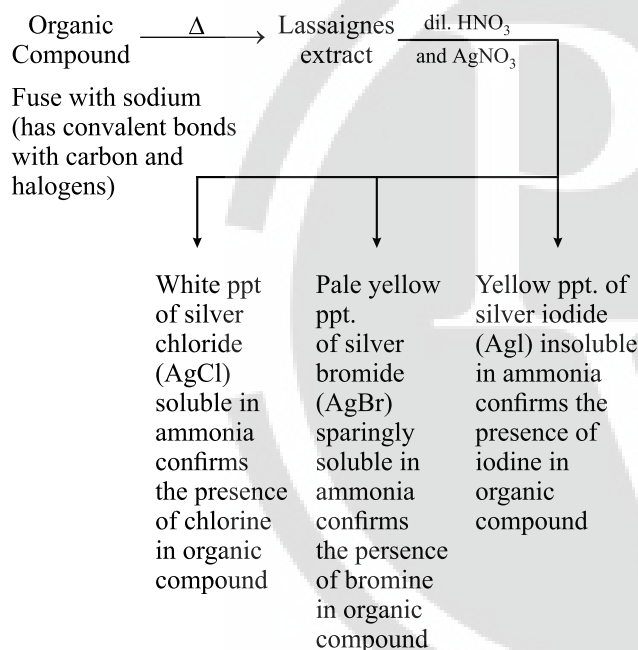
Lassaigne's extract is prepared to convert covalency of organic compound into electrovalency by fusing with Na.



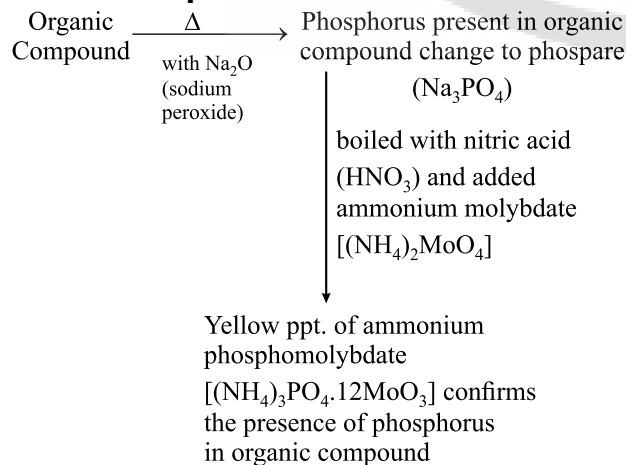
Detection of Halogen

Beilstein's test

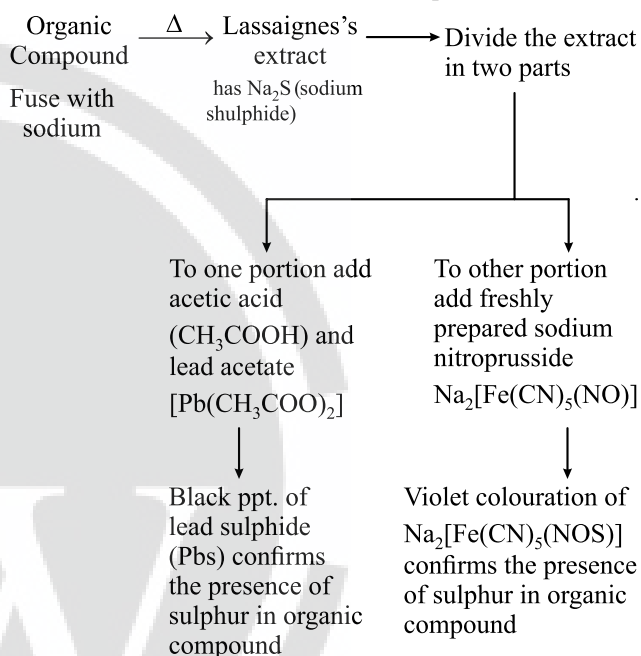
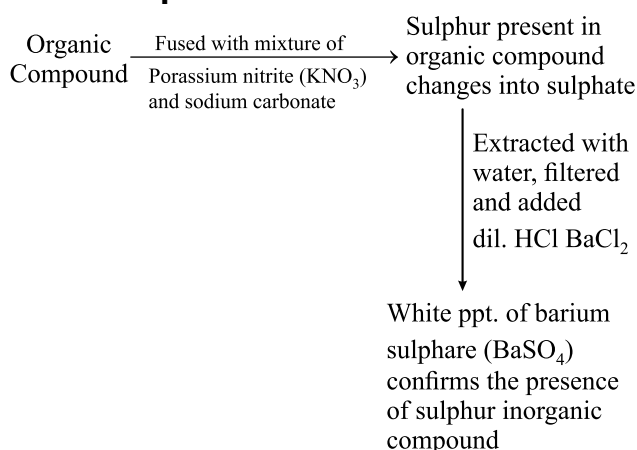
Organic compound $\xrightarrow[\text{Copper wire}]{\text{Heated over}}$ Green flame confirms the presence of halogen's in the organic compound



Detection of Phosphorus



Detection of Sulphur



Detection of Oxygen

Presence of oxygen in organic compound is detected by testing for functional group containing oxygen e.g. alcohol ($-\text{OH}$), aldehyde ($-\text{CHO}$), ketone (RCOR), carboxylic acid ($-\text{COOH}$), ester ($-\text{COOR}$) and nitro ($-\text{NO}_2$).

Quantitative Analysis

$$\%C = \frac{12}{44} \times \frac{\text{wt. of } \text{CO}_2}{\text{Wt. of org. compound}}$$

$$\%H = \frac{2}{18} \times \frac{\text{wt. of } \text{H}_2\text{O}}{\text{Wt. of org. compound}} \times 100$$

$$\%N = \frac{28}{22400} \times \frac{\text{Vol. of } \text{N}_2 \text{ at STP}}{\text{Wt. of org. comp.}} \times 100$$

or

$$\%N = \frac{\text{Volume of } \text{N}_2 \text{ at STP}}{8 \times \text{Wt. of org. compound}} \quad (\text{Duma's method})$$

$$(\%N) = \frac{1.4 \times N \times V}{\text{wt. of org. compound}} \text{ (Kjeldhal's method)}$$

$$\%X = \frac{\text{At. wt. of X}}{\text{Mol. wt. AgX}} \times \frac{\text{Wt. of AgX}}{\text{wt. of org. compound}}$$

$$\%S = \frac{32}{233} \times \frac{\text{Wt. BaSO}_4}{\text{wt. of org. compound}} \times 100$$

$$\%P = \frac{62}{222} \times \frac{\text{Mass of Mg}_2\text{P}_2\text{O}_7}{\text{Wt. of org. compound}} \times 100$$

